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ASSIGNMENT DAY 1

Overview of Hardware Components.

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TASK 1.

❖ Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.

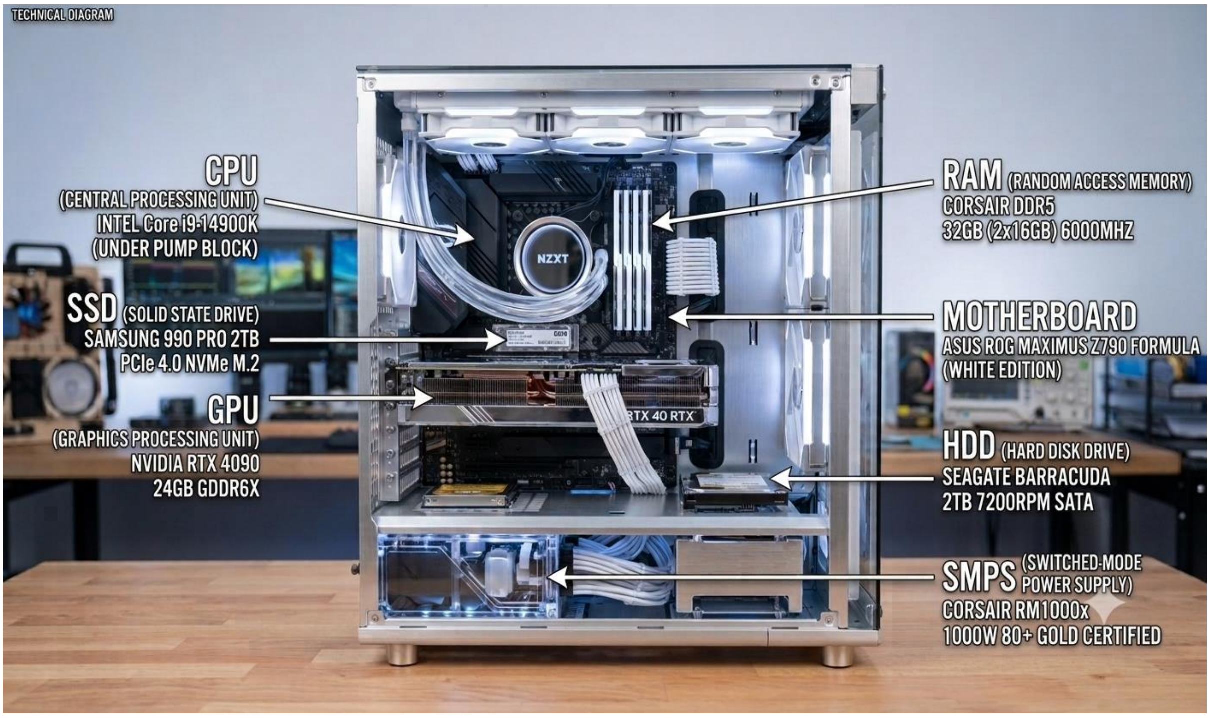

FIGURE 1 : LABELED SCHEMATIC OF A DESKTOP PC ARCHITECTURE.

Technical Description of Core Components

✓ Motherboard: The main circuit board of the computer. It connects all hardware components and allows them to communicate smoothly with each other.

✓ CPU (Central Processing Unit): The brain of the computer. It performs calculations, processes instructions, and runs the operating system and applications.

✓ GPU (Graphics Processing Unit): The component that handles graphics. It improves gaming, video editing, and other graphics intensive tasks by reducing the CPU’s workload.

- ✓ RAM (Random Access Memory): The computer's temporary memory, It stores the data and programs currently in use, helping the system run smoothly and multitask efficiently.
- ✓ SSD (Solid State Drive): A fast storage device that uses flash memory. It provides quick boot times and loads applications much faster.
- ✓ HDD (Hard Disk Drive): A storage device with a large capacity. It uses spinning disks to store files, backups, and other large amounts of data at a lower cost

TASK 2.

❖ Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases. Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.)

➢ While both HDDs and SSDs are used for permanent data storage, they use different technologies. The table below compares them based on their technology, speed, durability, and common uses.

✓ **Comparison: HDD vs. SSD**

Parameter	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Technology	Uses spinning magnetic disks and moving read/write heads.	Electronic, Uses NAND flash memory chips with zero moving parts.

Speed & Performance	Slower, Typical speed is 100–150 MB/s, so boot and loading times are longer.	Much faster. Speed ranges from 500 MB/s to over 7000 MB/s, giving quick boot and loading times.
Durability (Shock Resistance)	Less durable, it can be damaged if dropped because it has moving parts.	More durable, It is shock resistant because it has no moving parts.
Typical Use Cases	Used for storing large files, backups, CCTV recordings, and external storage.	Used as the main drive in laptops, gaming PCs, and computers that need fast performance.

- ✓ Practical Application: Many modern computers use both SSD and HDD together, The SSD is used for the operating system and important software to improve speed, while the HDD is used to store large files and backups at a lower cost.

TASK 3.

❖ Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

- ✓ CPU (Central Processing Unit): In games like PUBG or Free Fire, the CPU manages game logic, player actions, and game physics. It also tracks players positions and sends this information to the GPU so the game is displayed correctly.

- ✓ RAM (Random Access Memory): RAM stores the game data currently in use, such as maps, character models, and sound files. This helps the CPU and GPU access data quickly, reducing lag and keeping the game running smoothly.

- ✓ GPU (Graphics Processing Unit): The GPU renders the game's graphics, including 3D environments, textures, and lighting. It provides smooth visuals and higher frame rates (FPS) for a better gaming experience.

- ✓ **Summary:** In simple words, the CPU manages the game's tasks, RAM stores active game data for quick access, and the GPU displays smooth graphics on the screen.

TASK 4.

❖ Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

- ➢ To run the latest version of FIFA smoothly, I would upgrade the following hardware components:

  - ✓ Graphics Card (GPU): I would upgrade the GPU because it improves graphics quality, increases FPS (frames per second), and provides smoother gameplay.

  - ✓ RAM: I would increase the RAM to at least 16 GB. More RAM helps the game load faster and allows smooth multitasking without lag.

- ✓ Storage (SSD): I would replace the HDD with an SSD. An SSD reduces game loading time, speeds up Windows, and improves overall system performance.

- ✓ Processor (CPU): If my current processor is old, I would upgrade it to a newer and faster CPU. A better processor improves game performance and prevents bottlenecks.

- ✓ Power Supply (SMPS): If the new GPU and CPU require more power, I would upgrade the SMPS to provide stable and reliable power to all components.

- ✓ Upgrade Strategy: I would first upgrade the GPU and SSD for the biggest performance improvement. Next, I would upgrade the RAM and CPU if needed. Finally, I would ensure that the SMPS can provide enough power to support the upgraded components.